

Expression of orbital speed of a satellite.

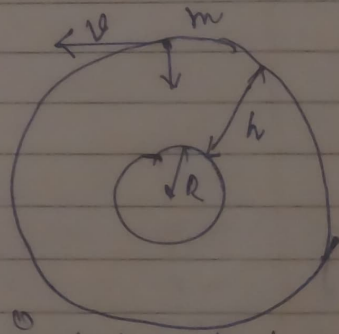
Let m , M and R be the mass & mean radius of earth respectively.

Consider a satellite of mass ' m ' revolving around earth at a height h . The necessary centripetal force for the satellite is provided by gravitational force.

Centripetal force = Gravitational force

$$\Rightarrow \frac{m v^2}{(R+h)} = \frac{G M m}{(R+h)^2}$$

$$v_0 = \sqrt{\frac{GM}{(R+h)}}$$



Orbital speed of a satellite is independent of mass.

Orbital speed of satellite decreases with increase in height.